Rosario Scalise

rosarioscalise.com
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ML Researcher with extensive experience on full-stack physical platforms.

Education

PhD, Machine Learning, University of Washington
Advising: Dr. Byron Boots, Dr. Abhishek Gupta, Dr. Dieter Fox, Dr. Yejin Choi

MS, Robotics, Carnegie Mellon University, SCS
Advising: Dr. Siddhartha S. Srinivasa, Dr. Sanjiban Choudhury, Dr. Stefanos Nikolaidis

BS, Electrical Engineering, University of Connecticut
Advising: Dr. Shalabh Gupta

BS, Computer Engineering, University of Connecticut
Minors: Mathematics, Digital Arts

Technical Skills

Languages: Python, MATLAB, LaTeX, bash **Libraries**: OpenCV, PyTorch, JAX, SciPy stack
Dev: Unix-like OS family, git, CI/CD, HPC for ML (SLURM), python: [pytest, uv/pip/poetry, setuptools]

Experience

UW Machine Learning & Robotics — PhD Student
University of Washington

Researching machine learning and optimization applied to

Technical Contracting / Internships:

Nvidia: Led the development of the first Digital Twin for a federal proving ground designed to be used as the basis of a DARPA program. Empirically studying long-horizon reasoning & planning in embodied AI.

Toyota Research Institute [link]: Co-authored novel application of OfflineRL to the problem of using suboptimal data to robustify imitation-based policies.

Personal Robotics Laboratory — Graduate Research Assistant
Carnegie Mellon University

- Researched human-robot interaction, enabling effective communication in human-robot collaborative manipulation tasks. Supported lab hardware and software infrastructure. Developed HERB 3.0

Electric Space Propulsion Lab — Research Staff
UCLA

- Automated experimental fixtures for applied plasma physics research: 2-DoF linear actuators in cryo-vacuum chambers, high-voltage dielectric barrier discharge (DBD) devices, laser-induced fluorescence (LIF) spectroscopy.

General Electric — Summer Intern (x3)
Plainville, CT

- Supported Entellisys LV switchgear: a wireless sensor network of industrial circuit breakers coordinated via a centralized computing stack with real-time guarantees. Developed field testkit and Human-Machine Interface.

Publications

J1 **R. Scalise**, S. Li, H. Admoni, S. Rosenthal, and S.S. Srinivasa. (2018). Natural language instructions for human-robot collaborative manipulation. *The International Journal of Robotics Research (IJRR).* [github]

C1 T. Han, Y. Bao, B. Mehta, G. Guo, A. Vishwakarma, E. Kang, S. Jung, **R. Scalise**, J. Zhou, B. Xu, & B. Boots. (2025). Model predictive adversarial imitation learning for planning from observation [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2507.21533

C2 K. Huang*, **R. Scalise***, C. Winston, A. Agrawal, Y. Zhang, R. Baijal, M. Grotz, B. Boots, B. Burchfiel, M. Itkina, P. Shah, & A. Gupta. (2025). Using non-expert data to robustify imitation learning via offline reinforcement learning [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2510.19495

C3 M. G. Castro, S. Rajagopal, D. Gorbatov, M. Schmittle, R. Baijal, O. Zhang, **R. Scalise**, S. Talia, E. Romig, C. de Melo, B. Boots, & A. Gupta. (2025). VAMOS: A hierarchical vision-language-action model for capability-modulated and steerable navigation [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2510.20818

C4 M. Schmittle, R. Baijal, N. Hatch, **R. Scalise**, M. G. Castro, S. Talia, K. Khetarpal, B. Boots, & S. Srinivasa. (2025). Long range navigator (LRN): Extending robot planning horizons beyond metric maps *Conference on Robot Learning (CoRL).*

C5 Y. Yang, G. Shi, C. Lin, X. Meng, **R. Scalise**, M. G. Castro, W. Yu, T. Zhang, D. Zhao, J. Tan, & B. Boots. (2025). Agile continuous jumping in discontinuous terrains. In 2025 IEEE International Conference on Robotics and Automation (ICRA)

C6 T. Han, P. Shah, S. Rajagopaal, Y. Bao, S. K. Jung, S. S. Talia, G. Guo, B. Xu, B. Mehta, E. Romig, **R. Scalise**, B. Boots. (2025). Demonstrating Wheeled Lab: Modern Sim2Real for Low-cost, Open-source Wheeled Robotics *Conference on Robot Learning (CoRL).* [pdf]

C7 A. Nanavati, E.K. Gordon, T. Faulkner, B. Zhu, T. Schrenk, J. Ko, A. Kashyap, R. Karim, H. Bolotski, **R. Scalise**, H. Song, Y. Qu, M. Cakmak, S.S. Srinivasa. (2025). Lessons Learned from Designing and Evaluating a Robot-Assisted Feeding System for Out-of-Lab Use In *Conference on Human Robot Interaction (HRI).* [pdf] **Best Systems Paper Award Nominee**

C8 **R. Scalise**, E. Caglar, B. Boots, C.C. Kessens. (2024). Toward Self-Righting and Recovery in the Wild: Challenges and Benchmarks In *International Conference on Robotics & Automation (ICRA).* [pdf]

C9 ..., **R. Scalise**, ... (2024). Open x-embodiment: Robotic learning datasets and rt-x models In *International Conference on Robotics & Automation (ICRA).* **Best Paper Award** [pdf]

C10 A. Khazatsky*, K. Pertsch*, ..., **R. Scalise**, ..., C. Finn (2024). DROID: A large-scale in-the-wild robot manipulation dataset In *Robotics: Science & Systems (RSS).* [pdf]

C11 **R. Scalise**, A. Mandalika, B. Hou, S. Choudhury, and S.S. Srinivasa. (2023). GuILD: Guided Incremental Local Densification for Accelerated Sampling-based Motion Planning In *International Conference on Robotics & Automation (ICRA).* [pdf]

C12 A. Lambert, B. Hou, **R. Scalise**, S.S. Srinivasa, and B. Boots. (2022). Stein Variational Probabilistic Roadmaps. In *International Conference on Robotics & Automation (ICRA).* [arxiv]

C13 T. Bhattacharjee, E. Gordon, **R. Scalise,** , M. Cabrera, A. Caspi, M. Cakmak, and S.S. Srinivasa. (2020). Is more autonomy always better? Exploring preferences of users with mobility impairments in robot-assisted feeding. In *International Conference on (HRI).* **Best Demonstration Award @ NeurIPS** [pdf]

C14 S.S. Srinivasa, P. Lancaster, ..., **R. Scalise**, ... (2019). MuSHR: A low-cost, open-source robotic racecar for education and research [arxiv]

C15 **R. Scalise**, J. Thomason, Y. Bisk, and S.S. Srinivasa. (2019). Improving Robot Success Detection using Static Object Data. In *International Conference on Intelligent Robots and Systems (IROS).* [arxiv]

C16 S. Li*, **R. Scalise***, H. Admoni, S. Rosenthal, and S.S. Srinivasa. (2017). Evaluating critical points in trajectories. In *Robot and Human Interactive Communication (RO-MAN).*

C17 **R. Scalise***, S. Li*, H. Admoni, S Rosenthal, and S.S. Srinivasa. (2016). Spatial references and perspective in natural language instructions for collaborative manipulation. In *Robot and Human Interactive Communication (RO-MAN)*.

W1 O. Zheng, Q. Peng, **R. Scalise**, B. Boots (2024). Parental Guidance: Efficient Lifelong Learning through Evolutionary Distillation

In *Evolutionary Robotics Workshop at Conference on Robot Learning (CoRL)*. **Best Paper Award**

W2 **R. Scalise**, S. Rosenthal, and S.S. Srinivasa. (2017). Natural Language Explanations in Human-Collaborative Systems. In *Proceedings of the Pioneers Workshop at (HRI)*.

W3 S. Li*, **R. Scalise***, S. Rosenthal, and S.S. Srinivasa. (2016). Perspective in Natural Language Instructions for Collaborative Manipulation. In *Workshop on Model Learning for Human-Robot Communication at (RSS)*.

Honors & Awards

Oak Ridge Associated Universities Doctoral Research Fellowship	2021
NeurIPS Best Demonstration Award	2018
HRI Pioneer	2017
University Honors Graduate	2015
Babbidge Scholar	2015
IEEE Power & Energy Society Scholarship Recipient	2013, 2014, 2015
New England Scholar	2014
Dominick A. Pagano Scholarship in Computer Science & Engineering Recipient	2013

Teaching

16-669: Robot Autonomy — Graduate Teaching Assistant
Carnegie Mellon University — Fall 2017

Mentoring & Outreach

Tao Jin — Safe Control & Interaction for Assistive Manipulators (Now: PhD at CMU)	2019 → 2020
Venkatesh Varada — Lovelace: The Autonomy-enabled Wheelchair	2018 → 2019
Jeffrey Maxwell — HERBTalk: A Natural Language Pipeline	2017 → 2019
Yiren (Ramon) Qu — Streaming RGBD & Tactile Data Wirelessly (Now: Google)	2017 → 2020
Maha Alrashed — Crowdsourcing YCB Object Affordances	Summer 2018
Abdullah Albakry — Reverse-engineering an R-net Wheelchair	Summer 2018
K-12 Lab Demos	Monthly

Eta Kappa Nu (HKN) Chair — 2013 → 2015
UConn Beta Omega Chapter

IEEE Vice-chair — 2012 → 2013
UConn Chapter

Academic Service

Conference Organization

IROS — 2019
Session on Semantic Scene Understanding — Chair

Conference Refereeing

ICRA: 2022, 2023, 2024 **IROS**: 2019; **RO-MAN**: 2017; **HRI**: 2019, 2020